RELATIVE MAYER-VIETORIS LONG EXACT SEQUENCE FOR HIGHER DIRECT IMAGES OF A TWO-OPEN COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author carries the functoriality claim over from the absolute statement without proof, and the bracketed name Relative Mayer-Vietoris in the environment header is reproduced as written.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot: sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both referenced results live in the same chapter and are located at their \label lines inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Mayer-Vietoris for a two-open cover, in its degreewise form. Same chapter, Lemma labelled lemma-mayer-vietoris (source line 767, Stacks tag 01EB); invoked at proof line 842, and its proof is where the displayed degreewise short exact sequence was established. Supplies the degreewise exactness of the sequence of sections over f^{-1}(W), which is what makes the sequence of complexes of direct images exact.
- Cohomology of an open subspace is computed by restriction of an injective resolution. Same chapter, Lemma labelled lemma-cohomology-of-open (source line 496, Stacks tag 01E1); invoked at proof lines 844-846. Supplies that the restrictions of an injective resolution are injective resolutions of the restricted modules, which identifies the cohomology sheaves appearing in the long exact sequence with the higher direct images.

Source reference. Original lemma 01EB: sources/stacks-cohomology/pinned-original-source.tex, source lines 767–767. Same-chapter pointer invoked at proof line 842 for the degreewise short exact sequence of sections; the assembly of the sequence of complexes of direct images and the reading off of the long exact sequence are retained in the counted span.

Source reference. Original lemma 01E1: sources/stacks-cohomology/pinned-original-source.tex, source lines 496–496. Same-chapter pointer invoked at proof lines 844-846 to identify the restrictions of the injective resolution as injective resolutions; the passage from the exact sequence of complexes to the long exact sequence of higher direct images is retained in the counted span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1 (Relative Mayer-Vietoris)**.** *Let $f : X \to Y$ be a morphism of ringed spaces. Suppose that $X = U \cup V$ is a union of two open subsets. Denote $a = f|_U : U \to Y$, $b = f|_V : V \to Y$, and $c = f|_{U \cap V} : U \cap V \to Y$. For every $\mathcal{O}_X$-module $\mathcal{F}$ there exists a long exact sequence*

$$0 \to f_*\mathcal{F} \to a_*(\mathcal{F}|_U) \oplus b_*(\mathcal{F}|_V) \to c_*(\mathcal{F}|_{U \cap V}) \to R^1 f_*\mathcal{F} \to \ldots$$

This long exact sequence is functorial in $\mathcal{F}$.

Proof. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}$. We claim that we get a short exact sequence of complexes

$$0 \to f_*\mathcal{I}^\bullet \to a_*\mathcal{I}^\bullet|_U \oplus b_*\mathcal{I}^\bullet|_V \to c_*\mathcal{I}^\bullet|_{U \cap V} \to 0.$$

Namely, for any open $W \subset Y$, and for any $n \geq 0$ the corresponding sequence of groups of sections over W

$$0 \to \mathcal{I}^n(f^{-1}(W)) \to \mathcal{I}^n(U \cap f^{-1}(W)) \oplus \mathcal{I}^n(V \cap f^{-1}(W)) \to \mathcal{I}^n(U \cap V \cap f^{-1}(W)) \to 0$$

was shown to be short exact in the proof of Lemma 01EB. The lemma follows by taking cohomology sheaves and using the fact that $\mathcal{I}^\bullet|_U$ is an injective resolution of $\mathcal{F}|_U$ and similarly for $\mathcal{I}^\bullet|_V$, $\mathcal{I}^\bullet|_{U \cap V}$ see Lemma 01E1. □